Classical Field Theory — Exercise Sheet 10

Residual gauges and charged tails

Questions only

Write $\Box = \partial_t^2 - \Delta$. Use $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \partial_t\mathbf{A}$, $\mathbf{B} = \operatorname{curl}\mathbf{A}$ and $\ell = -(\partial_t\phi_{\mathrm{em}} + \operatorname{div}\mathbf{A})$. Lecture 10's theorem assumes a smooth conserved source of compact *spacetime* support and compatible compact smooth potential data. Problems 2–3 explicitly separate a charged background outside that class.

Exercise 10.1 (All residual shifts of Cauchy data). Let a smooth Lorenz-gauge Maxwell potential have initial data $(a_\phi, b_\phi, a_A, b_A)$ as in Lecture 10, with a fixed source ρ. Let χ be the smooth wave with arbitrary real compact smooth data $\chi(0) = p$, $\chi_t(0) = q$. Set $\phi'_{\mathrm{em}} = \phi_{\mathrm{em}} - \chi_t$, $\mathbf{A}' = \mathbf{A} + \nabla\chi$.

(a) Find the increments of all four potential data. Verify directly that both compatibility expressions $b_\phi + \operatorname{div} a_A$ and $-\Delta a_\phi - \operatorname{div} b_A - \rho(0)$ are unchanged. Compare the two initial electromagnetic fields.

(b) Prove that fixing all four potential data rules out every nonzero residual gauge wave in this compact-data class. Contrast this with fixing only the electromagnetic fields. If compactness of the gauge data is dropped, explain why a constant χ is physically trivial.

Exercise 10.2 (A smooth charge with a Coulomb tail). Choose a radial nonnegative $b \in C_c^\infty(\mathbb{R}^3)$ with $\int b\,\mathrm{d}^3x = 1$ and support in $\overline{B_R(0)}$, $R > 0$. Fix real $Q \neq 0$ and prescribe the static source $\rho_Q(x) = Qb(x)$, $\mathbf{J} = 0$ for all time. For $r > 0$ define, writing $\rho_Q(s)$ for its radial profile,

$$M_Q(r) = 4\pi \int_0^r s^2 \rho_Q(s)\mathrm{d}s, \quad E_r(r) = \frac{M_Q(r)}{4\pi r^2}, \quad \phi_Q(r) = \int_r^\infty E_r(s)\mathrm{d}s.$$

(a) Prove that $E_Q(x) = E_r(|x|)x/|x|$ and $\phi_Q(|x|)$ extend smoothly across zero. Prove Gauss's law there as well as away from zero. Show that $(\phi_{\mathrm{em}}, \mathbf{A}) = (\phi_Q, 0)$ solves Maxwell in Lorenz gauge.

(b) Determine the exact potential and field for $r \geq R$. Prove $\phi_Q \notin L^2$, hence $\phi_Q \notin H^1$, while $\nabla\phi_Q \in L^2$ and the field energy is finite. Calculate the electric energy outside B_R.

(c) Identify the failed hypotheses of Lecture 10's theorem. Why does conservation alone not force $Q = 0$ for this source?

Exercise 10.3 (An affine charged evolution class). Keep the background of Exercise 10.2. Let $(\rho_1, J_1) \in C_c^\infty(\mathbb{R}^{1+3}; \mathbb{R} \times \mathbb{R}^3)$ be conserved. Give compact smooth perturbation data $(a_{1\phi}, b_{1\phi}, a_{1A}, b_{1A})$ satisfying

$$b_{1\phi} + \operatorname{div} a_{1A} = 0, \qquad -\Delta a_{1\phi} - \operatorname{div} b_{1A} = \rho_1(0).$$

Construct a smooth Lorenz-gauge solution for total source $(\rho_Q + \rho_1, J_1)$ and total initial data obtained by adding the static background. Prove its total charge is Q, its electric and magnetic fields have finite energy on each slice, and $\phi_{\mathrm{em}}(t) \in \phi_Q + H^1(\mathbb{R}^3)$ while $\phi_{\mathrm{em}}(t) \notin H^1$. State the uniqueness conclusion actually supplied by Lecture 10. Explain why this construction does not put the charged background in a Schwartz canonical phase space.

Exercise 10.4 (Local potentials with no global real one-form). On the oriented unit sphere, with area form $\sin\theta \mathrm{d}\theta \wedge \mathrm{d}\varphi$, let $g_{\mathrm{mag}} \neq 0$ be real. On $U_N = S^2 \setminus \{\text{south pole}\}$ and $U_S = S^2 \setminus \{\text{north pole}\}$, respectively, take

$$A_N = g_{\mathrm{mag}}(1 - \cos\theta)\mathrm{d}\varphi, \qquad A_S = -g_{\mathrm{mag}}(1 + \cos\theta)\mathrm{d}\varphi.$$

These are spatial real one-form potentials; the task uses exterior calculus from *Topology and Geometry*, Part I, not bundle theory as a prerequisite.

(a) Verify smoothness at each included pole, compute the common curvature $F = \mathrm{d}A_N = \mathrm{d}A_S$, and find its total flux.

(b) Find a gauge function on a simply connected azimuthal sector in the overlap. Prove that there is no single-valued real gauge function for this shift on the whole overlap, and no globally smooth real one-form A on S^2 with $\mathrm{d}A = F$.

The interpretation by nontrivial bundles and single-valued charged transition phases, including the allowed flux quantization, is a later topic: *Topology and Geometry*, Lectures 20–22. No bundle construction or quantization theorem is assumed in this exercise.